Statement of Purpose

I am interested in developing practical methods that make machine learning (ML) systems **robust**, especially to naturally occurring distribution changes in the environment such that they can be deployed **reliably** in the real world at **large-scale**.

I have broadly contributed towards this goal during my undergraduate at **IIT Delhi** and a productive tenure later at **Microsoft Research (MSR)** as a Research Fellow. Since then, I have published 3 papers in conferences like **CVPR**, **ECML**, **MLSys** and submitted one to **ICLR**, gaining invaluable skills in conducting research. Below, I describe how few of my past research experiences involving empirical approaches to learn robust classifiers and large-scale ML influenced my interests over time and what were my learnings.

Robust training with missing labels. In medical and web domains, often times, the observed training data distribution might not *follow* the distribution we want to learn about. This is manifested in the form of mislabeled data points, or more specifically *systematic* missing labels due to inherent biases in the policy or user clicks from which we curate the dataset. For instance, a policy/user tends to predict/click popular labels only. In tasks such as **eXtreme multi-label Classification (XC)**, where only a few labels from an enormous label space (reaching orders of millions) are relevant per data point, obtaining all the relevant labels for a data point is infeasible. Thus, datasets here suffer the same fate [1]. Below, I describe two approaches to learn robust extreme classifiers[1] that I worked on at MSR.

1. Using inductive biases. These datasets usually have a *long-tailed* distribution, where *majority* of labels have limited training points, which is further exacerbated due to missing labels. This is a major problem in applications such as recommendation, where these tail or rare labels provide niche and informative results for users. Extreme classifiers are prone to overfit to these rare labels, leading to poor generalization on majority label space. The objective was to address the performance gap on rare labels. I was advised by **Dr. Manik Varma** in this effort.

I observed siamese-style encoders[2] on the contrary perform well on rare labels by aligning semantically similar items together due to information sharing among labels, that is representations of "car" and "vehicle" would be close. However, their overall performance is poor due to under-fitting on data-rich popular or head labels. Looking for something that is *best of both worlds*, I noticed past works [2] have tried ensemble approaches of siamese encoder and extreme classifiers, but they suffered from *misalgined* prediction scores. I effectively managed to sidestep this by combining them during training itself. I came up with a simple distillation framework to supervise extreme classifiers using relevance probability scores of the siamese encoder amongst labels and data points. I empirically demonstrated huge improvements (**upto 5% absolute**) in rare label accuracies across benchmark datasets [3], without compromising already excellent head label accuracies.

Classifiers trained using this framework were deployed on Microsoft's Bing search engine for sponsored search, reinforcing its practical applicability on large-scale. This work is under review at **ICLR 2024** [4] where it has garnered positive feedback (5,6,6,8 / 10 points). Working on this, I learnt how *good* inductive biases (semantically similar items close) in the form of supervision from even *low* accuracy models (siamese encoders) can help in generalization.

2. Using data-centric approaches. Unfortunately, above approach is limited since there are abundant relevant missing pairs that have no textual or semantic similarity, for instance a data point "ferrari" is relevant to its category "expensive cars" or "luxury vehicle". In fact, the success of extreme classifiers is because they are not limited to the this semantic notion of relevance due to independent parameters for every label [5]. Past works [6] used heuristic approaches to mine pseudo-labels representative of missing labels via auxiliary sources of information (hyper-link graphs). However, they suffer from these mined labels being not *task-specific*. My ongoing research project advised by **Dr. Amit Sharma** and **Dr. Manik Varma** aims to solve this problem at its root cause.

Initially, I used Large Language Models (LLMs) to generate relevant pseudo-labels for data points, but they fail to scale for even few thousand points. Further, efficient smaller LMs lack the parametric knowledge to generate diverse pseudo-labels. For instance, a data point like "m118dw printer" has a relevant category "hewlett packard" that smaller LMs might just not *know* [7]. However, I observed that data points usually have abundant associated information but in an uncurated form. For instance, search queries have webpage titles (and webpage content) associated with them, products names have product descriptions etc. Therefore, I proposed to use a smaller LM (can scale to million data points) to generate task-specific pseudo-labels for a data point using this uncurated information. The smaller LM is aligned/supervised using a LLM like GPT4 for a specific task accordingly. These generated unsupervised but task specific pseudo-labels for data points can now be used to supervise efficient models.

This simple method efficiently scaled to 5 million data points and generated new relevant pairs, increasing available training data by 2x for a Wiki-category[3] dataset [3]. While the work is still in progress, this method can potentially learn robust representations which generalize to Out-Of-Distribution (OOD) labels due to the diversity of the generated pairs. Most importantly, I learnt that *simple* data-centric methods can go a long way in learning good representations.

[1] Extreme classifiers consist of a neural network encoder followed by a massive one-vs-all linear layer

[2] siamese encoders are transformer networks trained in a contrastive manner that encode both points and labels using common weights

[3] data points are Wikipedia titles, labels are categories

Reliable ML. During my undergrad at IIT Delhi, I worked with **Prof. Chetan Arora** on the problem of calibration in deep learning. Neural networks are calibrated when their predictions of an event align with the actual probability of that event occurring. This is extremely important in safety-critical applications such as *medical diagnosis* or *self-driving cars* where uncertainty estimates (probabilistic predictions) allow ML systems to establish trust and to defer to humans when they are unsure. Training classifiers using hard one-hot labels leads to overfitting and miscalibration [8, 9]. Consequently, training on soft full label distribution that reflect human uncertainty in labels leads to robust and reliable classifiers [10]. Since humans cannot provide full label distribution for every image, I realized that using pretrained models could be one potential way to provide label distributions for training classifiers. Using this insight, I used a pretrained but calibrated classifier to provide *accurate* uncertainty estimates for an image in the form its probabilistic predictions to supervise another classifier. This simple approach distills not only more accurate, but calibrated students, outperforming or on par with current baselines. I empirically show that we can only distill calibrated classifiers from calibrated teachers only. Interestingly, we can distill calibrated classifiers using smaller teachers too, as long as they are calibrated. With the help of a fellow undergrad, I conducted exhaustive experiments across datasets and demonstrated superior calibration. We submitted this work to **CVPR 2024**. This work followed naturally from my previous project, again with Prof. Chetan Arora that got accepted to **CVPR 2022** [11] as an **oral presentation** (*top 4.2% papers*), where we devised a simple loss that acts as an *engineered* label distribution. This reinforced the importance of good quality labels, and supervision from other ML models can be an effective way to approximate it.

Large-scale ML. Being in an industrial research lab offered me the unique opportunity to work with huge scales of architectures and datasets where performing even *trivial* operations on these would mean hours (or even days) of compute. I quickly adapted to these magnitudes and learned to write efficient code and algorithms that could scale, picking up valuable skills in **distributed** and **parallel** computing, especially on GPUs.

End-to-End training of large-scale XC models, that is training the encoder and the massive one-vs-all linear layer together, was previously thought to be infeasible in literature [2, 12]. I contributed to the research project towards making this end-to-end training practical and efficient, reducing training time by 15x, where I was advised by **Dr. Ramchandran Ramjee** and **Dr. Manik Varma**. I helped in implementation of the project and proposed a simple augmentation fix to learn better classifiers. I conducted exhaustive experiments demonstrating vanilla end-to-end training outperforms complicated modular approaches [2, 5] by upto 5% absolute points. This second-author work was received warmly at **MLSys 2023** [13].

I collaborated with data scientists across different product groups within Microsoft to actively deploy my research, learning valuable engineering skills and good code-practices on the way. I applied the distillation framework described previously [4] on a real world large-scale sponsored search dataset collected using Bing click logs containing close to 80 million labels. This deployment led to 25% increase in offline metrics and significant increase in clickthrough-rates and user satisfaction online. This made me realize that *simple* ideas can work better than complicated ones on large-scale datasets.

Future directions. My past experiences informed my research interests and made me realize the importance of developing practical methods. Therefore during graduate studies, I would like to use both theoretical and empirical approaches towards developing practical methods to handle distribution changes in the wild, especially at large scales. A possible direction where my previous research experience can help is exploring the **role of data** in robustness. One question directly relevant to my current work: Can we develop algorithms to learn task-specific representations that can generalize to hard to characterize natural shifts, leveraging abundant uncurated or unlabeled data? Another direction could be the **role of foundational models** in robustness: a possible question, how can we effectively exploit their general purpose representations, learnt from naturally occurring diverse data, in training task-specific models that can generalize when deployed in ever-changing real world? Additionally on a slight tangent, I'd also be interested in empirically understanding fundamental properties of ML models and use these learnings to develop better methods towards distributional changes. Further, I would like to complement my research work with extensive theoretical courses during my graduate studies to learn tools for theoretical research work.

At NYU. I am interested in working with **Prof. Pavel Izmailov** since his interests in developing practical methods to make ML models robust aligns with mine. Prof. Pavel's recent work [14] on training robust models in presence of spurious correlations using a simple data-centric finetuning on last layer falls close to one of the future directions I wish to pursue. Additionally, his past work [15] on understanding properties about neural networks empirically and build practical methods using these learnings [16, 17] (a calibration method) is something I am excited to explore further. Additionally, I would like to work with **Prof. He He**. Her interests in building ML models robust to biases in training dataset and being aware of "what they do not know" aligns with my current research project and past works on calibration and OOD detection [11, 18]. Further, I would also be interested in collaborating with **Prof. Saining Xie**. His recent work [19] on curating task-specific data during training falls close to my current research project as well as future directions I would want to pursue to learn *useful* and *robust* representations using abundant internet-scale data.

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